# Brickhub Intelligence — Assignment (Anurag)

Oct 12, 2026

## Overview

You will design and build a small intelligence tool that helps a real estate investor decide where in Pune to buy residential property, and you will test how far its answers can be trusted. We want to see how you frame the problem, make methodological choices under imperfect data, and communicate uncertainty honestly.

**The scenario.** A mid-sized investment firm wants to put capital into Pune residential real estate and needs a repeatable way to compare micro-markets (localities such as Kharadi, Baner or Hadapsar). The investment committee meets soon, and your tool should help them choose where to look first.

**Your task.** Build a prototype that scores 10 to 15 Pune micro-markets and recommends the top 3 for a residential investor, then stress-test your own ranking. Keep the scope small and finish it.

We are interested mostly in understanding your reasoning, your validation, and how plainly you state what the data cannot support.

## What you will build

Build a micro-market scoring and ranking tool for one city, one asset class and one investor type, with four analytical requirements on top. The limits keep the task small enough to finish in about six hours.

- **City:** Pune (you choose which 10 to 15 micro-markets to include, and tell us why)
- **Asset class:** Residential apartments only
- **Investor:** A buyer looking for a blend of capital appreciation and rental income over a 5 to 7 year hold
- **Output:** A ranked list of your micro-markets and a clear recommendation of the top 3

**The four analytical requirements.**

1. **Method choice.** Choose how you will turn indicators into a ranking (for example a weighted score, a statistical model or clustering), and justify why it suits this data size and this decision.

2. **Missing data.** Some of your data will be missing or unreliable. Handle this deliberately, and show how much your handling changes the ranking.
3. **Ranking stability.** Test how sensitive your top 3 is to your own assumptions, such as the weights or the data you imputed. Report which areas stay on top and which depend on your choices.
4. **What the data cannot support.** State clearly which conclusions you would not stand behind, given the size and quality of your data.

**Format.** Use whatever you are strongest in: Excel or Google Sheets, SQL, Power BI, Python, or R. We care that the logic is visible.

**Out of scope.** Do not build for other cities, commercial or industrial assets, or financial modelling such as IRR or loan structuring. If you think any of these matter, mention them in your limitations section instead.

## Deliverables

Submit four things. Write the design note first, before you start building, and do not rewrite it afterwards to match what you built.

| # | Deliverable | Format | Length |
|---|---|---|---|
| 1 | Design note | PDF or Word | 1 to 2 pages |
| 2 | Reproducible prototype and method note | Notebook or scripts, plus a short note | Method note up to 1 page |
| 3 | Analysis, recommendation and limitations | PDF or Word | 1 to 2 pages |
| 4 | Walkthrough | Live call with us | 15 minutes plus Q&A |

**1. Design note (before you build).** Answer these questions:

- What decision does this tool support, and who will use it?
- Which indicators will you use (for example demand, supply, price growth, rental yield, infrastructure, risk), and why those?
- What method will you use to combine them, and why is it appropriate for this much data?
- What data problems do you expect, and how do you plan to handle them?

**2. Prototype and method note.** Your deliverable should run from raw inputs to the final ranking. In the method note, explain your data sources, cleaning steps, scaling, how you handled missing values, how you chose weights or model parameters, and how you checked your work.

**3. Analysis, recommendation and limitations.** Give your top 3 micro-markets with a short reason for each. Show your stability and sensitivity results, ideally with one or two clear visuals. Then state which conclusions the data does not support.

**4. Walkthrough.** You will present your work for about 15 minutes. Expect questions on your method choices, your validation and one technical trade-off that you would explain to a non-technical partner.

## Data and tools

You source your own data from public information

- **Cite everything.** Record the source and date for each data point, and mark any value you estimated rather than found.
- **Gaps are expected.** If values are missing for some areas, do not hide it. Tell us how you handled it, and show how much it affects your ranking.
- **AI tools:** You may use AI assistants or search tools, but you must say where you used them.
- **Do not use** confidential or proprietary data from any current or past employer.

## Time and submission

Submit by end of day on Friday, 16th October 2026. Please do not spend more than that. We would rather see a focused, finished piece than an exhaustive one.

- If you run out of time, say what you would have done next. We value that over rushed work.
- Send your design note, code, method note and analysis by email as attachments or a shared link.
- Your design note should be a separate file, so we can see it was written before the build.
- If anything in the brief is unclear, make a reasonable assumption and write it down. You may also email us a clarifying question.

We will schedule your walkthrough after we receive your submission.

Thank you for your time and effort.